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METODA MIDPOINT COMPOZITA

Considerăm diviziunea echidistantă

$$a = x_0 < x_1 < \dots < x_{m-1} < x_m = b$$

$$\text{cu } h = x_i - x_{i-1}, i = \overline{1, m}, \quad x_i = a + ih, i = \overline{0, m};$$

$$\text{Notăm } x_{i+1/2} = \frac{1}{2} (x_i + x_{i+1}) \leftarrow \text{mijlocul intervalului } [x_i, x_{i+1}]$$

$$\text{Avem } \int_a^b f(x) dx = \sum_{i=0}^{m-1} \int_{x_i}^{x_{i+1}} f(x) dx$$

Aplicăm formula MIDPOINT pentru fiecare interval $[x_i, x_{i+1}]$

$$\int_{x_i}^{x_{i+1}} f(x) dx = (x_{i+1} - x_i) f(x_{i+1/2}) + \frac{(x_{i+1} - x_i)^3}{24} f''(\xi_i)$$

$$\text{unde } \xi_i \in [x_i, x_{i+1}]$$

$$\begin{aligned} \Rightarrow \int_a^b f(x) dx &= \sum_{i=0}^{m-1} \left[(x_{i+1} - x_i) f(x_{i+1/2}) + \frac{(x_{i+1} - x_i)^3}{24} f''(\xi_i) \right] = \\ &= h \sum_{i=0}^{m-1} f(x_{i+1/2}) + \sum_{i=0}^{m-1} \frac{h^3}{24} f''(\xi_i) \end{aligned}$$

Vom folosi teorema de medie discretă:

Fie $u \in C[a, b]$ și $d_0, d_1, \dots, d_m$ constante cu același semn. Atunci există $\xi \in [a, b]$ a.i.

$$\sum_{j=0}^m d_j u(x_j) = u(\xi) \sum_{j=0}^m d_j$$

Considerăm $u = f''$ și $d_0 = d_1 = \dots = d_{m-1} = 1$ ($m = m-1$)

$$\Rightarrow \exists \xi \in [a, b] \text{ a.i. } \sum_{j=0}^{m-1} f''(\xi_j) = f''(\xi) \sum_{j=0}^{m-1} d_j = m f''(\xi)$$

Obținem

$$\int_a^b f(x) dx = \frac{b-a}{m} \sum_{i=0}^{m-1} f(x_{i+1/2}) + \frac{(b-a)^3}{24m^2} f''(\xi)$$

$$\xi \in [a, b]$$

Mai exact, am înlocuit $h = \frac{b-a}{n}$ în

$$\int_a^b f(x) dx = h \sum_{i=0}^{n-1} f(x_{i+1/2}) + \sum_{i=0}^{n-1} \frac{h^3}{24} f''(\xi_i) =$$

$$= \frac{b-a}{n} \sum_{i=0}^{n-1} f(x_{i+1/2}) + \frac{(b-a)^3}{24n^3} \sum_{i=0}^{n-1} f''(\xi_i) =$$

$$= \frac{b-a}{n} \sum_{i=0}^{n-1} f(x_{i+1/2}) + \frac{(b-a)^3}{24n^3} \cdot n f''(\xi) = \frac{b-a}{n} \sum_{i=0}^{n-1} f(x_{i+1/2}) + \frac{b-a}{24n^2} f''(\xi)$$

METODA TRAPEZULUI COMPOZITĂ

Considerăm diviziunea echidistantă

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$$

cu $h = x_i - x_{i-1}$, $i = \overline{1, n}$, $x_i = a + ih$, $i = \overline{0, n}$, $h = \frac{b-a}{n}$

Avem $\int_a^b f(x) dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x) dx$

Aplicăm metoda trapezului pentru fiecare interval $[x_i, x_{i+1}]$

$$\int_{x_i}^{x_{i+1}} f(x) dx = \frac{x_{i+1} - x_i}{2} [f(x_i) + f(x_{i+1})] - \frac{(x_{i+1} - x_i)^3}{12} f''(\xi_i)$$

unde $\xi_i \in [x_i, x_{i+1}]$

$$\Rightarrow \int_a^b f(x) dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x) dx =$$

$$= \sum_{i=0}^{n-1} \left[\frac{x_{i+1} - x_i}{2} (f(x_i) + f(x_{i+1})) - \frac{(x_{i+1} - x_i)^3}{12} f''(\xi_i) \right] =$$

$$= \sum_{i=0}^{n-1} \frac{h}{2} (f(x_i) + f(x_{i+1})) - \sum_{i=0}^{n-1} \frac{h^3}{12} f''(\xi_i) =$$

$$= \frac{h}{2} [f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)] - \frac{h^3}{12} \sum_{i=0}^{n-1} f''(\xi_i) =$$

$$= \frac{h}{2} [f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)] - \frac{(b-a)^3}{12n^3} \sum_{i=0}^{n-1} f''(\xi_i) =$$

$$\begin{aligned}
 &= \frac{b-a}{2m} \left[f(x_0) + 2 \sum_{i=1}^{m-1} f(x_i) + f(x_m) \right] - \frac{(b-a)^3}{12m^3} f''(\xi) = \\
 &= \frac{b-a}{2m} \left[f(x_0) + 2 \sum_{i=1}^{m-1} f(x_i) + f(x_m) \right] - \frac{(b-a)^3}{12m^2} f''(\xi)
 \end{aligned}$$

$\xi \in [a, b]$

Obținem

$$\int_a^b f(x) dx = \frac{b-a}{2m} \left[f(x_0) + 2 \sum_{i=1}^{m-1} f(x_i) + f(x_m) \right] - \frac{(b-a)^3}{12m^2} f''(\xi)$$

$\xi \in [a, b]$

METODA SIMPSON COMPOZITĂ

Considerăm diviziunea echidistantă pentru $n=2m$ (par)

$$a = x_0 < x_1 < \dots < x_{2m-1} < x_{2m} = b$$

$$\text{cu } x_i - x_{i-1} = \frac{h}{2}, \quad i = \overline{1, 2m}, \quad x_i = a + i \frac{h}{2}$$

$i = \overline{0, 2m}$

$$h = \frac{b-a}{m}$$

$$\text{Avem } \int_a^b f(x) dx = \sum_{i=0}^{m-1} \int_{x_{2i}}^{x_{2i+2}} f(x) dx$$

Aplicăm metoda Simpson pentru fiecare interval $[x_{2i}, x_{2i+2}]$

$$\begin{aligned}
 \int_{x_{2i}}^{x_{2i+2}} f(x) dx &= \frac{x_{2i+2} - x_{2i}}{6} \left[f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2}) \right] - \\
 &\quad - \frac{(x_{2i+2} - x_{2i})^5}{2880} f^{(4)}(\xi_{2i})
 \end{aligned}$$

unde $\xi_{2i} \in [x_{2i}, x_{2i+2}]$

$$\begin{aligned}
 \Rightarrow \int_a^b f(x) dx &= \sum_{i=0}^{m-1} \int_{x_{2i}}^{x_{2i+2}} f(x) dx = \sum_{i=0}^{m-1} \left(\frac{x_{2i+2} - x_{2i}}{6} \left[f(x_{2i}) + \right. \right. \\
 &\quad \left. \left. + 4f(x_{2i+1}) + f(x_{2i+2}) \right] - \frac{(x_{2i+2} - x_{2i})^5}{2880} f^{(4)}(\xi_{2i}) \right) =
 \end{aligned}$$

$$\begin{aligned}
&= \frac{h}{6} \sum_{i=0}^{m-1} [f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})] - \\
&\quad - \frac{h^5}{2880} \sum_{i=0}^{m-1} f^{(4)}(\xi_{2i}) = \\
&= \frac{h}{6} [f(x_0) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + f(x_{2m})] - \\
&\quad - \frac{h^5}{2880} \sum_{i=0}^{m-1} f^{(4)}(\xi_{2i}) = \frac{h}{6} [f(x_0) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + \\
&\quad + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + f(x_{2m})] - \frac{h^5}{2880} m f^{(4)}(\xi) = \\
&= \frac{b-a}{6m} [f(x_0) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + f(x_{2m})] - \\
&\quad - \frac{(b-a)^5}{2880 m^5} m f^{(4)}(\xi)
\end{aligned}$$

Obținem:

$$\begin{aligned}
\int_a^b f(x) dx &= \frac{b-a}{6m} [f(x_0) + 2 \sum_{i=1}^{m-1} f(x_{2i}) + 4 \sum_{i=0}^{m-1} f(x_{2i+1}) + f(x_{2m})] - \\
&\quad - \frac{(b-a)^5}{2880 m^5} f^{(4)}(\xi) = \\
&\quad \xrightarrow{m=\frac{n}{2}} \frac{b-a}{3n} [f(x_0) + 2 \sum_{i=1}^{n/2-1} f(x_{2i}) + 4 \sum_{i=0}^{n/2-1} f(x_{2i+1}) + f(x_n)] - \\
&\quad - \frac{(b-a)^5}{180 n^5} f^{(4)}(\xi), \quad \xi \in [a, b]
\end{aligned}$$

Observație Pt metoda Simpson "n" trebuie să fie par ($n=2m$), iar metoda se aplică pentru fiecare interval de forma $[x_0, x_2]$, $[x_2, x_4]$, $[x_4, x_6]$, ...; punctele de indice impar $x_1, x_3, \dots, x_{n-1}$ se folosesc doar în cadrul "formulei de bază"

METODA SIMPSON $\frac{3}{8}$ = 18 = COMPOZITĂ

Considerăm diviziunea echidistantă pentru $n = 3k$
(multiplu de 3)

$$a = x_0 < x_1 < \dots < x_{3k-1} < x_{3k} = b$$

$$\text{cu } x_i - x_{i-1} = \frac{h}{3}, i = 1, 3k, x_i = a + i \frac{h}{3}, i = 0, 3k$$

$$h = \frac{b-a}{k}$$

$$\text{Avem } \int_a^b f(x) dx = \sum_{i=0}^{k-1} \int_{x_{3i}}^{x_{3i+3}} f(x) dx$$

Aplicăm metoda Simpson $\frac{3}{8}$ pt fiecare interval $[x_{3i}, x_{3i+3}]$

$$\int_{x_{3i}}^{x_{3i+3}} f(x) dx = \frac{x_{3i+3} - x_{3i}}{8} \left[f(x_{3i}) + 3f(x_{3i+1}) + 3f(x_{3i+2}) + f(x_{3i+3}) \right] - \frac{(x_{3i+3} - x_{3i})^5}{6480} f^{(4)}(\xi_{3i})$$

unde $\xi_{3i} \in [x_{3i}, x_{3i+3}]$

$$\Rightarrow \int_a^b f(x) dx = \sum_{i=0}^{k-1} \int_{x_{3i}}^{x_{3i+3}} f(x) dx =$$

$$= \sum_{i=0}^{k-1} \left(\frac{x_{3i+3} - x_{3i}}{8} \left[f(x_{3i}) + 3f(x_{3i+1}) + 3f(x_{3i+2}) + f(x_{3i+3}) \right] - \frac{(x_{3i+3} - x_{3i})^5}{6480} f^{(4)}(\xi_{3i}) \right) =$$

$$= \frac{h}{8} \sum_{i=0}^{k-1} \left[f(x_{3i}) + 3f(x_{3i+1}) + 3f(x_{3i+2}) + f(x_{3i+3}) \right] - \frac{h^5}{6480} \sum_{i=0}^{k-1} f^{(4)}(\xi_{3i}) =$$

$$\begin{aligned}
 &= \frac{h}{8} \left[f(x_0) + 3 \sum_{i=0}^{k-1} f(x_{3i+1}) + 3 \sum_{i=0}^{k-1} f(x_{3i+2}) + 2 \sum_{i=1}^{k-1} f(x_{3i}) + \right. \\
 &\quad \left. + f(x_{3k}) \right] - \frac{h^5}{6480} f^{(4)}(\xi) = \\
 &= \frac{h}{8} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \neq 3}}^{3k-1} f(x_i) + 2 \sum_{i=1}^{k-1} f(x_{3i}) + f(x_{3k}) \right] - \\
 &\quad - \frac{h^5}{6480} f^{(4)}(\xi) = \frac{b-a}{8k} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \neq 3}}^{3k-1} f(x_i) + \right. \\
 &\quad \left. + 2 \sum_{i=1}^{k-1} f(x_{3i}) + f(x_{3k}) \right] - \frac{(b-a)^5}{6480k^5} f^{(4)}(\xi)
 \end{aligned}$$

Obținem $i \neq 3 \rightarrow$ înseamnă că nu se divide la 3

$$\begin{aligned}
 \int_a^b f(x) dx &= \frac{b-a}{8k} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \neq 3}}^{3k-1} f(x_i) + 2 \sum_{i=1}^{k-1} f(x_{3i}) + \right. \\
 &\quad \left. + f(x_{3k}) \right] - \frac{(b-a)^5}{6480k^5} f^{(4)}(\xi) \quad \leftarrow k = \frac{n}{3}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3(b-a)}{8n} \left[f(x_0) + 3 \sum_{\substack{i=1 \\ i \neq 3}}^{n-1} f(x_i) + 2 \sum_{i=1}^{n/3-1} f(x_{3i}) + \right. \\
 &\quad \left. + f(x_n) \right] - \frac{(b-a)^5}{80n^5} f^{(4)}(\xi), \quad \xi \in [a, b]
 \end{aligned}$$

Pt metoda Simpson $3/8$ "n" trebuie să fie multiplu de 3 ($n=3k$), iar metoda se aplică pentru fiecare interval de forma $[x_0, x_3]$, $[x_3, x_6]$, $[x_6, x_9]$, ...; punctele de indexi care nu se divid la 3 ($i \neq 3$) $x_1, x_2, x_4, x_5, x_8, \dots, x_{n-1}$ se folosesc doar în cadrul "formulei de bază"